

GABLS1: SLD $\gamma=2$

GABLS1 stable surface-layer diffusivity, $\gamma=2$ | 23 September 2026. This matched nine-hour 32-cubed, 12.5 m WENO9 case changes only interior SLD stability strength. The same 400 m domain, seed 123, filtered wall, one-face support, 300 s filters, and scheme-native factor-one flux apply to $\gamma=0, 1, 2$, and 4. The fixed 1 m LES median is an intercomparison reference, not observations.

The 8-9 h vector shear at 12.5, 25, and 37.5 m is 0.09152, 0.06140, and 0.05235 per second at $\gamma=2$, versus 0.10038, 0.06666, and 0.05607 in the median profile. These values decrease with height and remove the pronounced 25 m dip of $\gamma=4$, whose first three shears are 0.12911, 0.03456, and 0.04001. $\gamma=1$ gives 0.07387, 0.07235, and 0.05350. $\gamma=2$ first-layer shear changed only from 0.08910 in 7-8 h to 0.09152 in 8-9 h.

Shear improvement does not imply improvement in all statistics. Peak resolved w -squared is 0.05588 square meters per second squared at $\gamma=2$ in 8-9 h, below $\gamma=1$ at 0.06676, $\gamma=4$ at 0.08575, and filtered no closure at 0.10125. At 18.75 m, w -cubed is +0.000937 cubic meters per second cubed, versus +0.005932 for $\gamma=4$ and +0.01629 with no closure. The $\gamma=2$ surface friction velocity is 0.27977 meters per second and heat flux is -0.010968 kelvin meters per second.

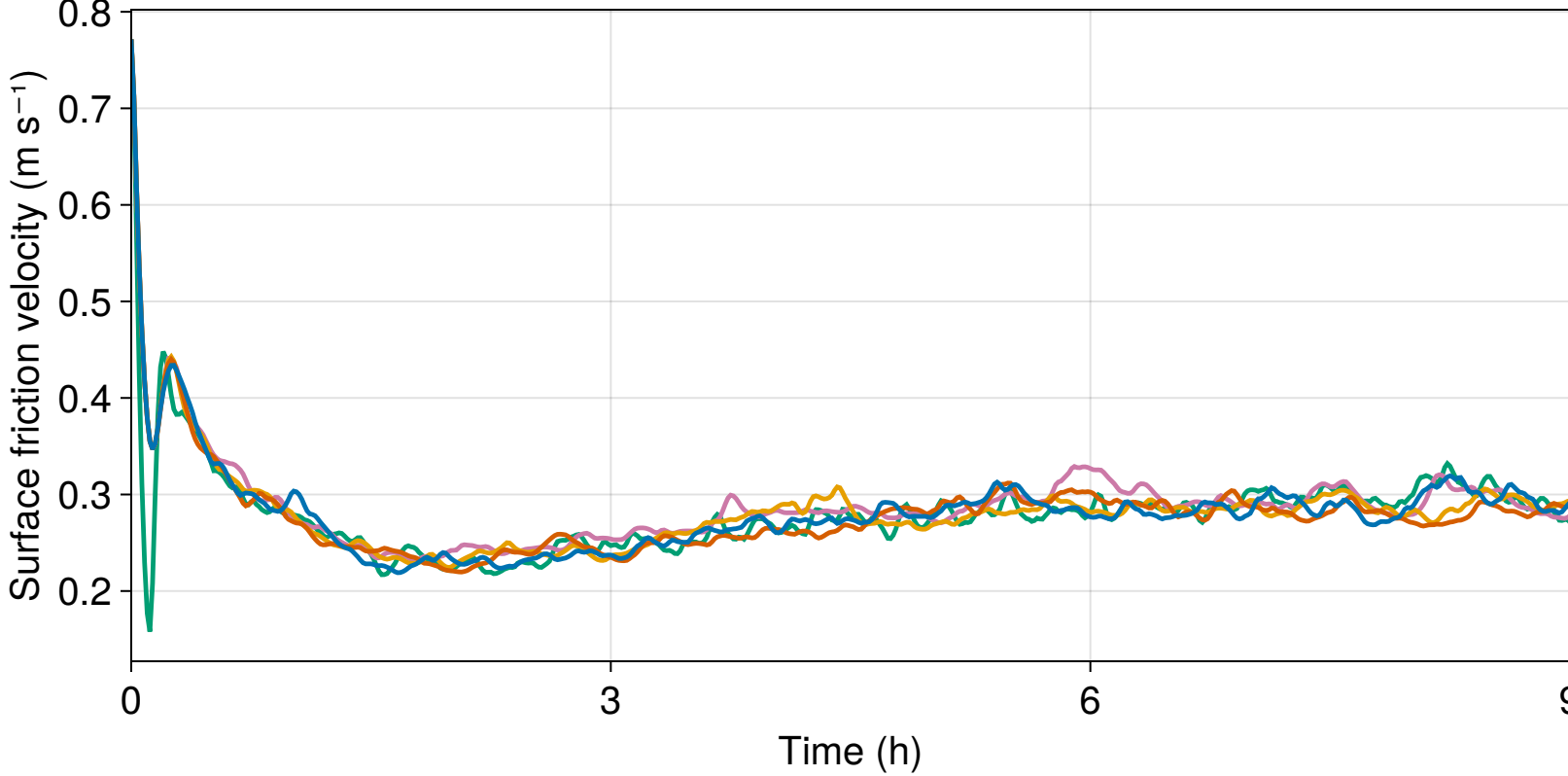
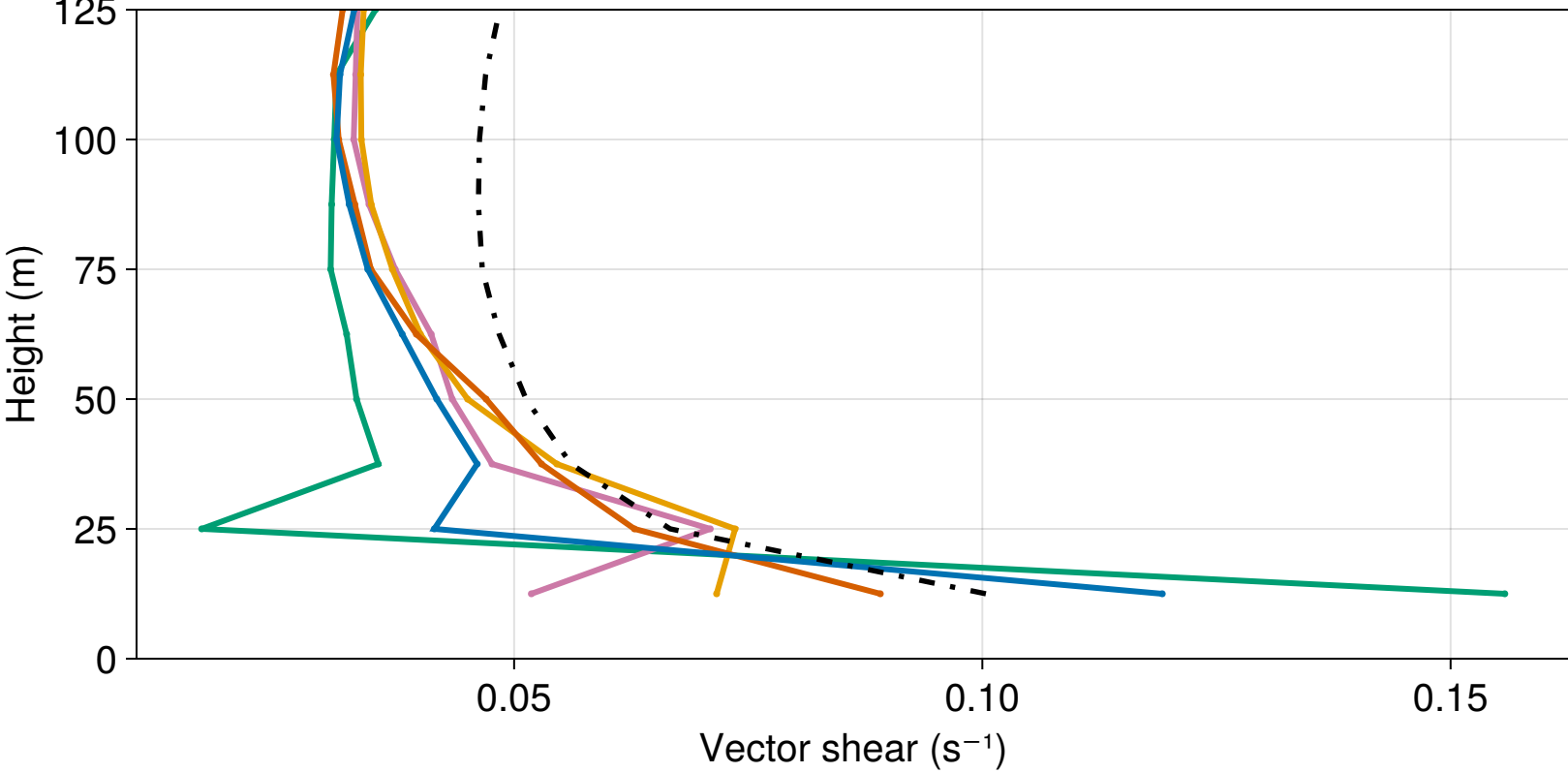
At the first interior face, $\gamma=2$ mean local $\phi_m=1.8930$ and $\phi_h=2.4512$ during 8-9 h; all saved columns stay stable and no upward-flux fallback occurs. Evolved momentum viscosity is 0.52303 square meters per second, between $\gamma=1$ at 0.82387 and $\gamma=4$ at 0.24517. Resolved $u-w$ is -0.021102 and SLD $u-w$ is -0.040393 square meters per second squared. The measured WENO numerical correction is -0.00069141 and is kept separate from covariance.

Validation: CPU smoke 7595 passed 48/48; H100 gate 7596 passed CUDA parity 34/34 and its 1800 s case 48/48. Science 7615_1 ended with durable exit zero and CASE_DONE at 32400 s. Independent Julia audits verified frozen source, paired initial-theta digest, exact schedules, 46 native-height profiles, 135 series, finite fields, filter evolution, density-consistent surface flux and scheme-native transport. Local inverse Obukhov length and similarity identities passed at 55 saved times. One seed and one coarse grid do not establish universal calibration; a paired 64-cubed study follows.

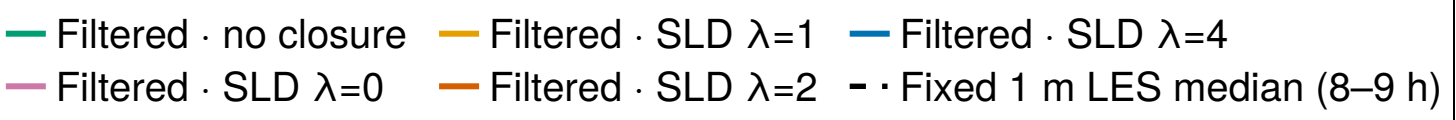
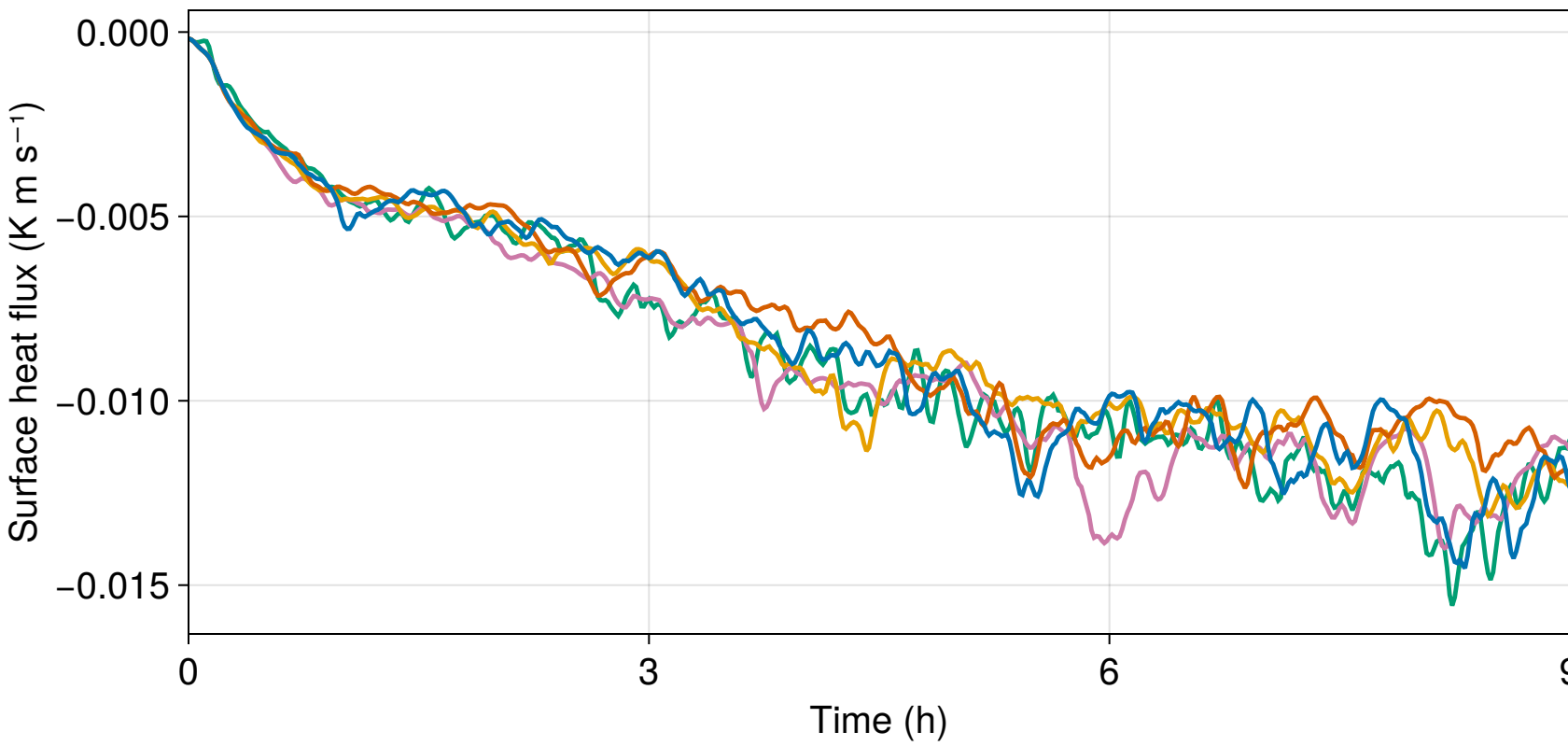
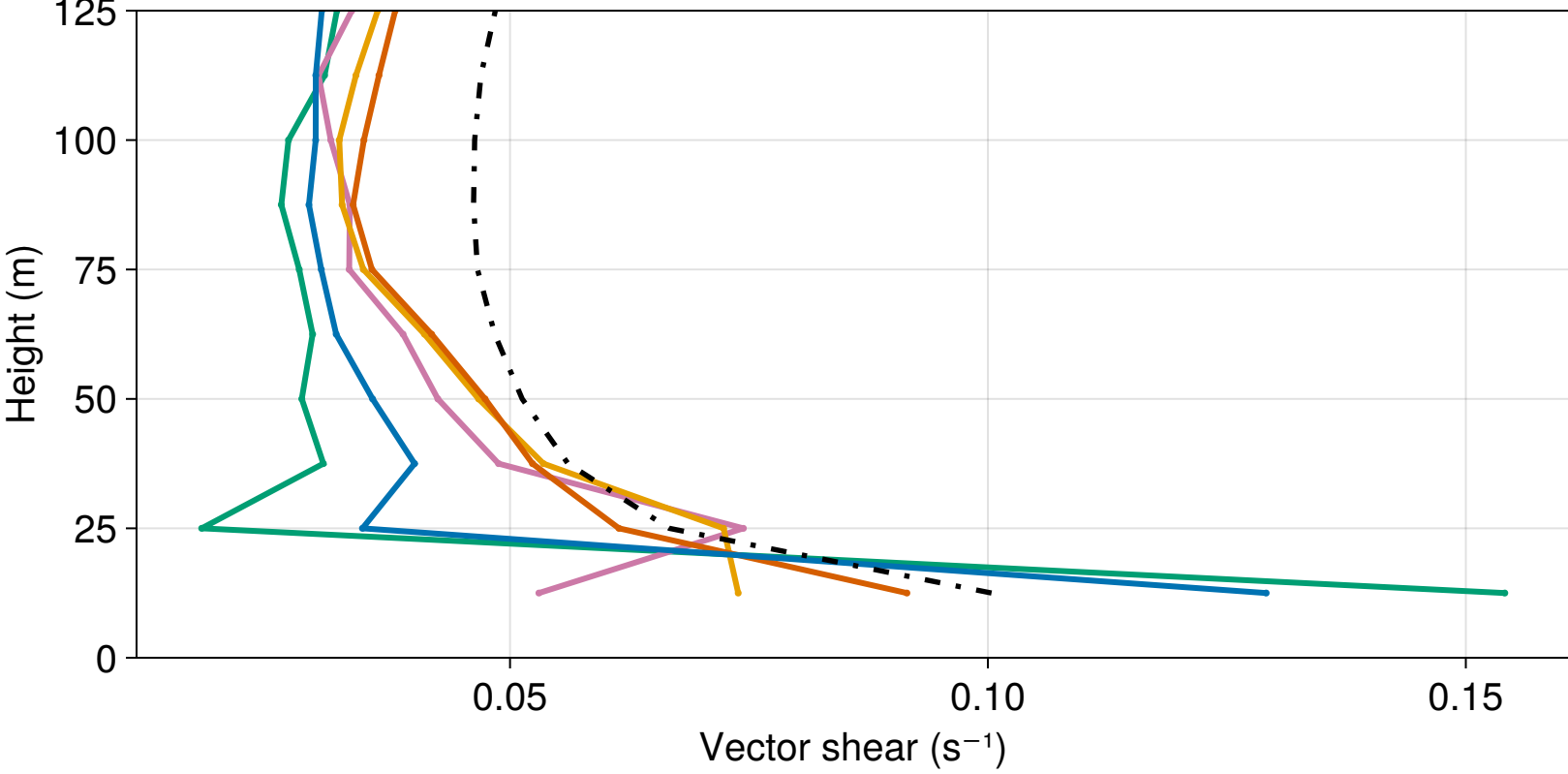
Mean-wind shear and wall flux

Shear between adjacent 12.5 m cell centers; fixed 1 m median sampled at the same heights

7–8 h

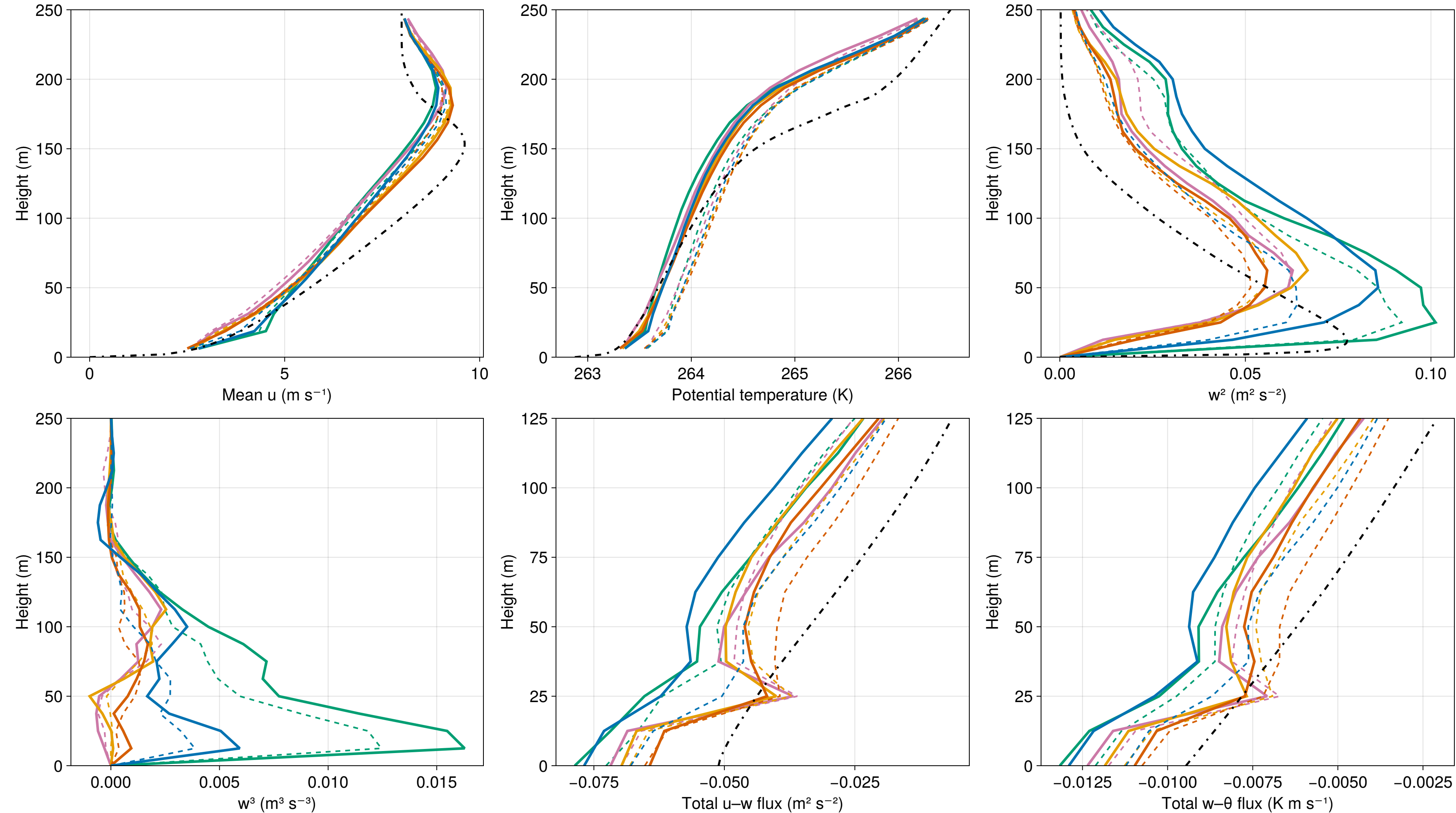


8–9 h



GABLS1: stable Monin–Obukhov correction of SLD

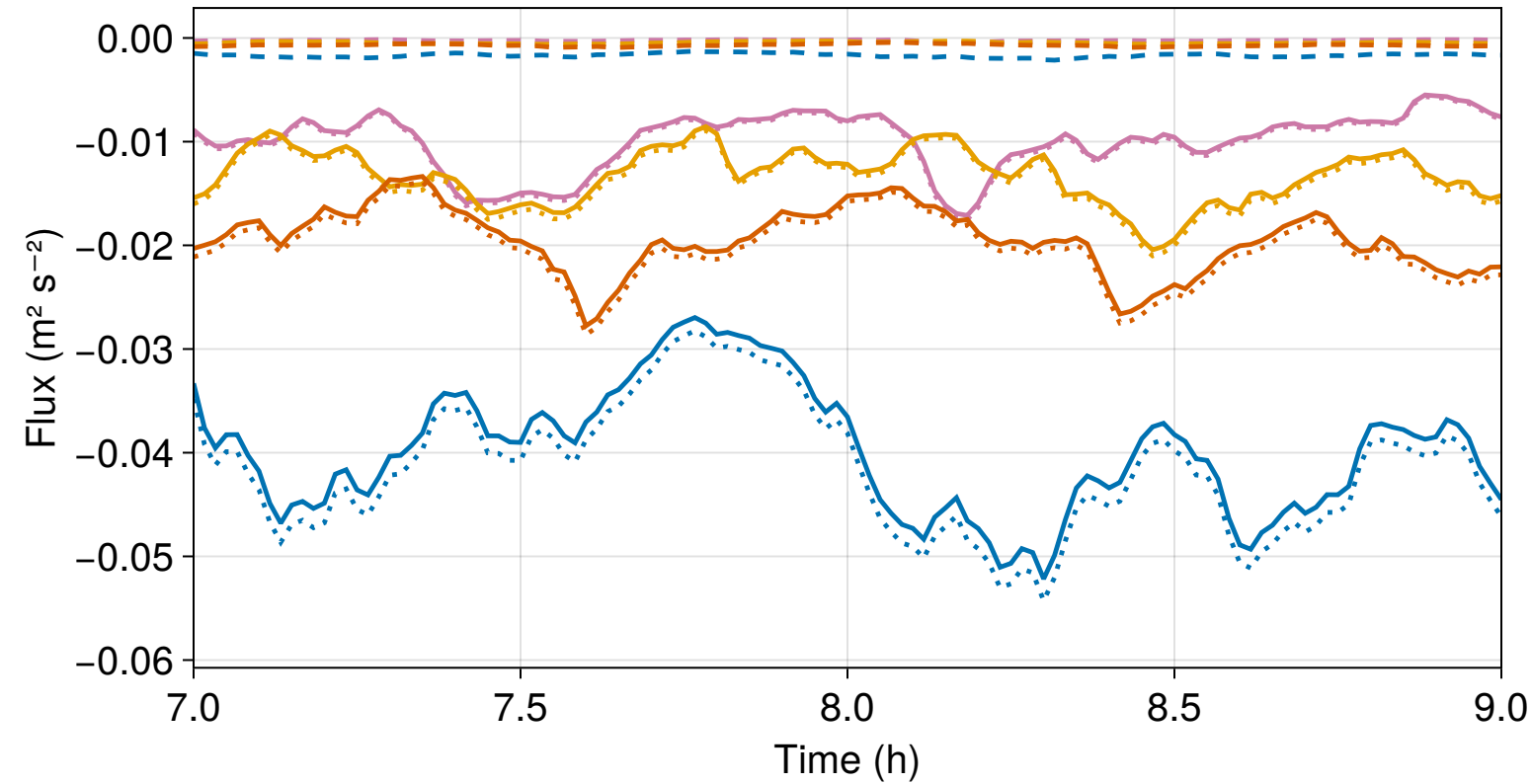
32³ · 12.5 m · WENO9 · filtered wall · scheme-native one-face SLD · $\lambda=0, 1, 2, 4$ · dashed 7–8 h, solid 8–9 h



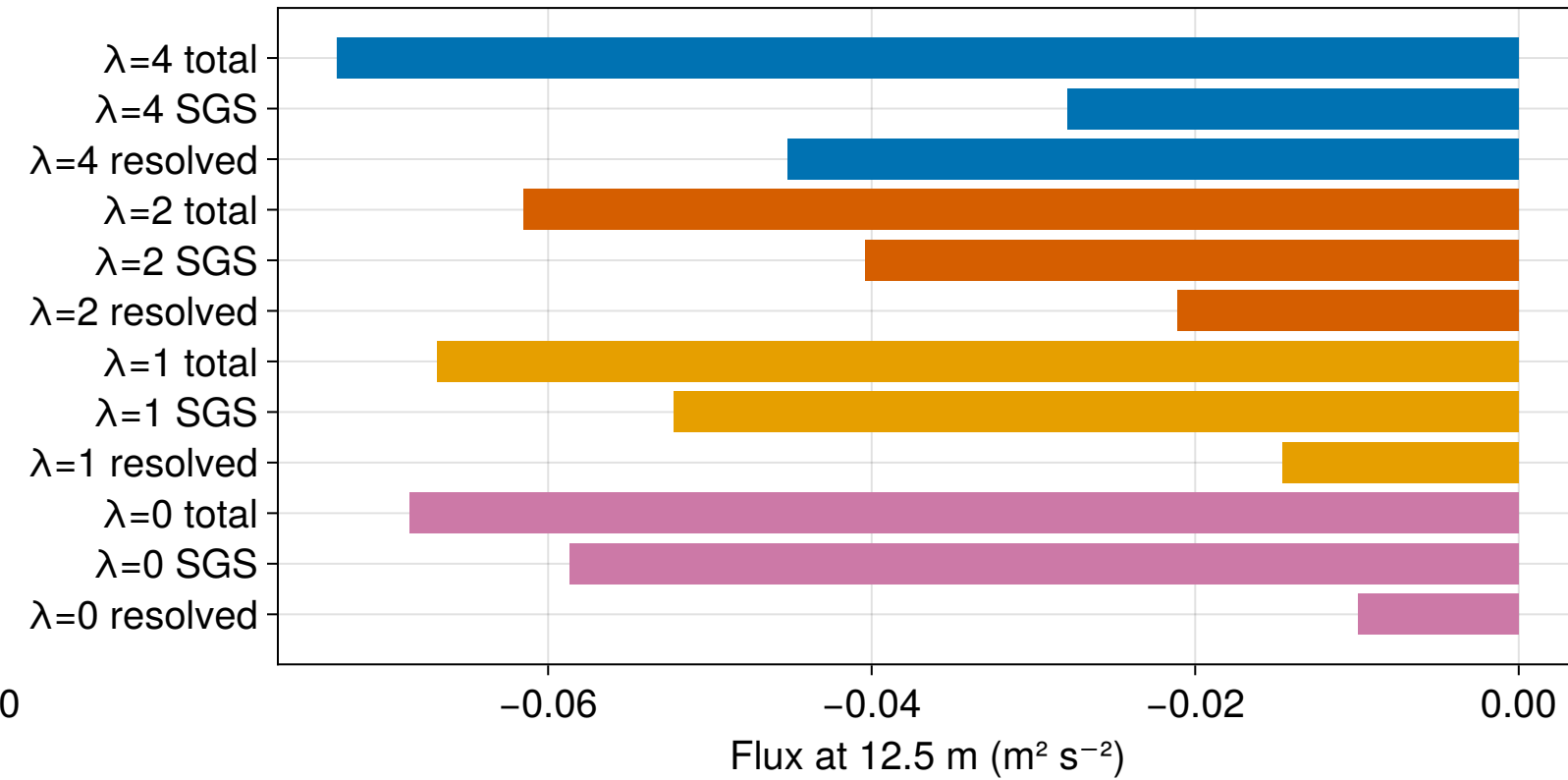
Scheme-native SLD: first-face transport partition

Reconstructed resolved = covariance + numerical correction; SGS is the SLD constitutive flux

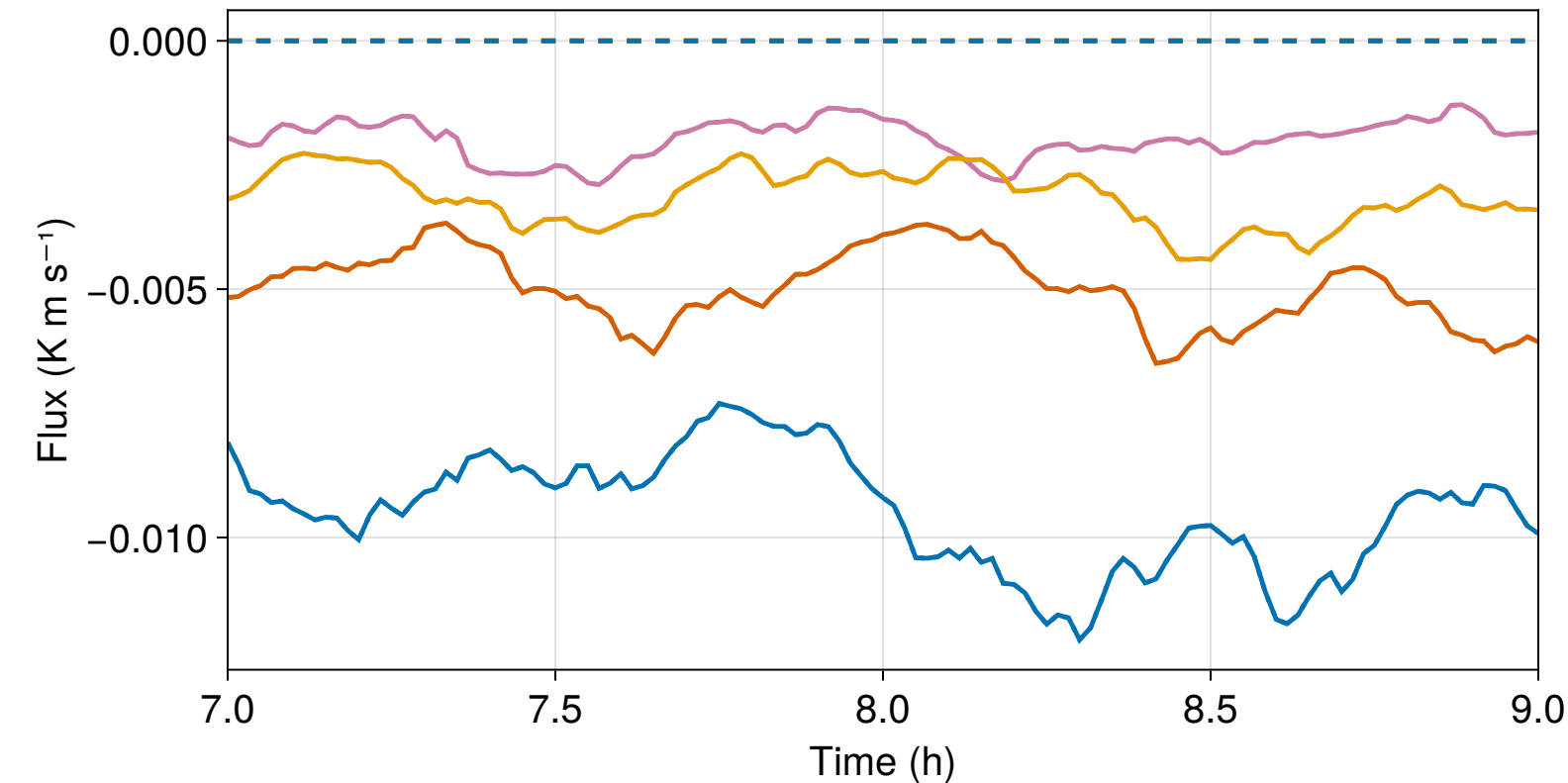
Momentum



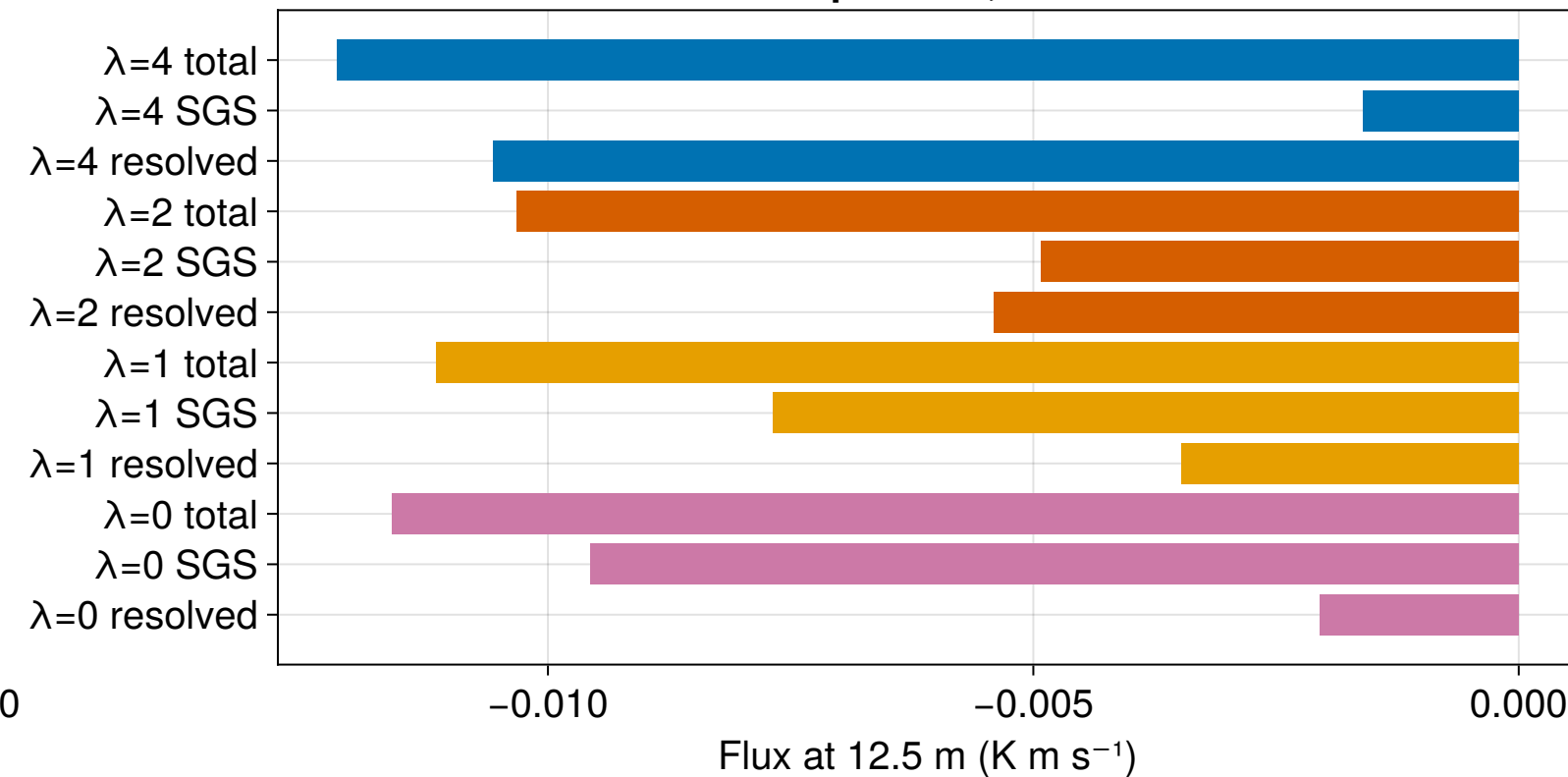
u-w partition, 8–9 h



Heat



w-θ partition, 8–9 h



— Filtered · SLD $\lambda=0$ covariance
 - - - Filtered · SLD $\lambda=0$ reconstructed
 — Filtered · SLD $\lambda=1$ covariance
 - - - Filtered · SLD $\lambda=1$ reconstructed
 — Filtered · SLD $\lambda=2$ covariance
 - - - Filtered · SLD $\lambda=2$ reconstructed
 — Filtered · SLD $\lambda=4$ covariance
 - - - Filtered · SLD $\lambda=4$ reconstructed

— Filtered · SLD $\lambda=0$ correction
 - - - Filtered · SLD $\lambda=0$ reconstructed
 — Filtered · SLD $\lambda=1$ covariance
 - - - Filtered · SLD $\lambda=1$ reconstructed
 — Filtered · SLD $\lambda=2$ covariance
 - - - Filtered · SLD $\lambda=2$ reconstructed
 — Filtered · SLD $\lambda=4$ covariance
 - - - Filtered · SLD $\lambda=4$ reconstructed

First-face SLD coefficients and local stability

Horizontal means of local quantities at $z = 12.5$ m; φ is never evaluated from a mean L

